

Phillip J. Macias

Computational Systems Modeler

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Summary

Computational systems modeler with 10+ years of experience designing and executing large-scale simulation frameworks in collaborative, deadline-driven environments. Experienced in distributed HPC workflows, high-dimensional parameter exploration, and uncertainty quantification and propagation. Contributed to simulation infrastructure for a NASA flagship mission.

Technical Expertise

- **Programming:** Python, C, Fortran, Bash | **HPC:** Slurm, distributed simulation campaigns, large-scale parameter sweeps
- **Modeling:** Monte Carlo methods, uncertainty quantification, sensitivity analysis
- **Data & Workflow:** Jupyter, NumPy, SciPy, Pandas, Matplotlib, Git, reproducible pipelines

Experience

University of California, Santa Cruz

2012 – 2026

Ph.D. Researcher (2012–2019); Postdoctoral Researcher (2020–2026)

Computational Modeling & Simulation

- Contributed modeling deliverables within a **35-member** Science Investigation Team for the **NASA Nancy Grace Roman Space Telescope**; core member of a **6-person** simulation subgroup.
- Designed and executed distributed simulation campaigns spanning **thousands of model realizations** on Slurm-managed HPC clusters.
- Explored **high-dimensional parameter spaces** to quantify system-level tradeoffs and uncertainty propagation.
- Debugged distributed pipeline failures and produced structured documentation for long-term reproducibility.
- Designed and ran large-scale **hydrodynamic simulations** using the FLASH AMR code; executed multi-day runs across **hundreds of CPU cores** to investigate nonlinear stability.

Mentorship & Leadership

- Mentored **15+** students (high school through graduate level) in computational modeling and research workflows.
- Led a paid remote research cohort of **five** students at The Thacher School, coordinating milestones and deliverables.

Research Output

Co-author of **20** peer-reviewed publications in astrophysics and computational modeling.

Education

Ph.D., Astronomy & Astrophysics — University of California, Santa Cruz

2019

M.S., Astronomy & Astrophysics — University of California, Santa Cruz

2015

B.S., Physics (Highest Honors) — University of California, Santa Barbara

2012